

Do Latent User Clusters Reveal Hidden Group Unfairness? An Evaluation with Neural Recommenders on LastFM and Yelp

Miguel Vieira Machado Pim
Universidade Federal do Espírito Santo
Vitória, Espírito Santo, Brasil
miguelmip02@gmail.com

Athus Cavalini
Universidade Federal do Espírito Santo
Vitória, Espírito Santo, Brasil
athus.cavalini@ifes.edu.br

Diego Roberto Colombo Dias
Universidade Federal do Espírito Santo
Vitória, Espírito Santo, Brasil
diego.dias@ufes.br

Hygo Freire Tagarro
Universidade Federal do Espírito Santo
Vitória, Espírito Santo, Brasil
hygo2025@gmail.com

Abstract

Fairness in recommender systems is an important concern, particularly when recommendation quality differs across user groups defined by demographic, behavioral, or social characteristics. This study investigates whether latent groups identified through unsupervised clustering reveal disparities beyond those observed through predefined user groupings. We evaluated two neural recommendation models, NeuMF and MultiVAE, on the LastFM-360K and Yelp datasets, using larger sampled user-item spaces and comparing predefined user characteristics with groups identified through K-Means and Agglomerative Clustering. Group unfairness was evaluated using R_{grp} , which measures differences in prediction losses across groups. The results show that clustering-based groups do not systematically produce the highest group unfairness. In LastFM, Age yielded the highest disparity for NeuMF ($R_{grp} = 0.00031714$), whereas Activity and the clustering-based partitions produced the largest values for MultiVAE, around 0.0037. In Yelp, Friends produced the highest unfairness for both NeuMF (0.06349989) and MultiVAE (0.07795636). Moreover, clustering-based partitions frequently resembled the Activity grouping in both group composition and fairness outcomes. These findings indicate that latent groups should not automatically be interpreted as evidence of previously hidden unfairness and highlight the importance of examining their composition alongside observable user characteristics.

CCS Concepts

• **Information systems** → **Recommender systems**; *Collaborative filtering*; • **Computing methodologies** → **Cluster analysis**; *Neural networks*.

Keywords

Recommender Systems, Group Fairness, Latent Groups, User Clustering

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1 Introduction

Recommender systems have become an important mechanism for helping users navigate large collections of products, services, and digital content. Recommender systems are commonly evaluated using metrics that summarize overall recommendation quality [17]. However, such measures may conceal substantial differences in the quality experienced by different users, motivating the study of fairness in recommender systems [3, 21]. From the consumer perspective, group fairness concerns the fair treatment of users belonging to different groups [20].

A central aspect of group fairness evaluation is therefore how the user population is divided. Previous studies commonly define groups using observable demographic or behavioral characteristics, such as gender, age, location, and activity [7, 15, 4]. Such partitions make it possible to identify disparities that would otherwise be hidden by aggregate performance measures. However, they restrict the analysis to characteristics selected in advance. More complex disparities may instead be associated with combinations of user characteristics that are not directly represented by any single predefined grouping.

This limitation motivates the investigation of data-driven user groups. Recent work explored this direction by applying hierarchical clustering to demographic and behavioral user attributes and evaluating the resulting groups through differences in recommendation error [4]. Using MovieLens and Amazon Books, the study reported that clustering-based groups frequently exposed substantial group unfairness, in some cases exceeding the disparities observed for predefined groups. These findings suggest that clustering may complement traditional fairness evaluations by revealing structures that are not apparent when users are partitioned according to one observable characteristic at a time.

It remains unclear, however, whether this behavior persists under substantially different recommendation settings. The previous experiments were conducted on sampled matrices containing 300

users and 1,000 items and considered traditional collaborative filtering approaches, including ALS, NMF, and KNN [4]. In this work, we investigate the same underlying question in two new recommendation domains, using larger sampled user–item spaces and neural recommendation models. Rather than assuming that clustering necessarily reveals hidden unfairness, we examine whether data-driven groups actually expose disparities beyond those already observed through predefined user characteristics.

We conduct experiments on LastFM-360K and the Yelp Open Dataset, sampling 10,000 users and 10,000 items from each dataset after preprocessing. We evaluate two neural recommendation architectures with distinct modeling assumptions: Neural Matrix Factorization (NeuMF) [9] and Variotinal Autoencoders (MultiVAE) [13]. Users are analyzed using predefined demographic, behavioral, and social characteristics available in each dataset, as well as groups identified independently through K-Means and Agglomerative Clustering. Following previous work [16, 4], group unfairness is measured from differences between group-level prediction losses. Overall recommendation performance is evaluated separately using NDCG@10.

Based on this setting, we address the following research questions:

- **RQ1:** How does group unfairness vary across predefined user characteristics and neural recommendation models in the LastFM and Yelp domains?
- **RQ2:** Do user groups identified through unsupervised clustering expose disparities beyond those revealed by predefined user groupings?

Our results show that clustering-based groups do not systematically produce the largest group unfairness. In LastFM, Age exposes the highest disparity for NeuMF, whereas the Activity and clustering-based partitions produce the largest values for MultiVAE. In Yelp, Friends produces the highest group unfairness for both recommendation models. Moreover, the clustering solutions in both datasets closely resemble the Activity partition in group sizes and fairness outcomes. These observations suggest that a clustering solution should not be interpreted as evidence of previously hidden unfairness solely because it forms internally coherent groups; the composition of the resulting groups and their relationship with observable user characteristics must also be considered.

The main contributions of this work are:

- an empirical evaluation of consumer-side group fairness with two neural recommendation models across two recommendation domains and larger sampled user–item spaces than those considered in the motivating study;
- a systematic comparison between predefined user groupings and data-driven groups obtained through K-Means and Agglomerative Clustering; and
- empirical evidence that clustering-based groups do not systematically expose greater disparities and may closely reflect structures already represented by observable behavioral characteristics.

2 Related Work

Fairness in recommender systems has been studied from multiple perspectives, including fairness toward users, item providers, and

other stakeholders [3, 21]. In this work, we focus on consumer-side fairness, where the subjects of the fairness analysis are the users receiving recommendations. Within consumer-side fairness, group fairness concerns whether recommendation quality is comparable across different user groups [5, 6, 8]. In this work, we adopt the loss-based formulation of [16], which quantifies disparities through differences in group-level prediction error.

A common strategy for studying consumer-side group fairness is to partition users according to observable characteristics. Demographic differences in recommendation effectiveness have been reported for attributes such as gender and age, demonstrating that aggregate evaluation can conceal substantial variation across user populations [7]. Behavioral characteristics can also define relevant groups. For example, activity-based analyses have shown that recommendation quality and fairness may differ between highly active and less active users [15]. These findings indicate that conclusions about fairness depend not only on the recommendation algorithm but also on the user characteristic used to define the groups under evaluation.

Restricting the analysis to predefined attributes, however, may overlook disparities associated with more complex user structures. In the broader fairness literature, subgroup fairness research has demonstrated that satisfying fairness criteria for groups defined by single predefined attributes does not necessarily prevent disparities in structured subgroups defined by combinations of attributes [11]. Similar concerns arise in recommender systems, where fairness has often been evaluated with respect to individual sensitive attributes rather than multiple fairness dimensions [19].

User fairness has been studied across different recommendation strategies [12], including neural collaborative filtering models extended to reduce disparities associated with predefined protected attributes, such as gender-based disparities in career recommendation [10]. These studies show that fairness concerns remain relevant across different recommendation architectures. More broadly, fairness-aware recommender studies commonly define user groups a priori from known sensitive or observable attributes [14, 16, 18]. Our setting differs by using unsupervised clustering to investigate whether additional groups with unequal recommendation quality emerge beyond these predefined partitions.

The work most directly related to ours is the study of group fairness based on hierarchical user clustering proposed in [4]. The authors evaluated ALS, NMF, and KNN on MovieLens and Amazon Books and compared predefined demographic and behavioral groups with groups obtained through Agglomerative Clustering. Their results showed that the hierarchical clustering configuration frequently produced high values of group unfairness, motivating the use of clustering as a mechanism for identifying disparities that may not be evident from predefined attributes alone.

Our work revisits this observation under a different experimental setting. We analyze LastFM and Yelp using NeuMF and MultiVAE on sampled matrices of 10,000 users and 10,000 items, and we consider both K-Means and Agglomerative Clustering. More importantly, we do not assume that a data-driven partition necessarily represents a new fairness dimension. We compare its fairness outcomes with those of the observable characteristics used to describe the same user population. In this way, the study investigates not only whether clustering can expose group disparities, but also whether

the discovered groups provide information beyond that already captured by predefined user characteristics.

3 Methodology

This section describes the methodology adopted in this study. It is organized into the following components: (i) datasets (ii) data preprocessing, (iii) recommendation models, (iv) hyperparameter optimization, (v) user grouping, and (vi) group fairness evaluation.

3.1 Datasets

We used two public datasets from different recommendation domains: **LastFM-360K** [2] and the **Yelp Open Dataset**¹. LastFM contains approximately 17 million user-artist interactions involving about 359,000 users and 294,000 artists. Each interaction records the number of times a user listened to a particular artist. The Yelp dataset contains approximately 7 million user-business ratings involving about 2 million users and 150,000 businesses, with ratings given on a 1-to-5 scale.

The datasets were selected to evaluate the proposed methodology across domains with different interaction characteristics and user metadata. LastFM represents listening behavior through interaction counts, whereas Yelp represents explicit user evaluations through ratings. This distinction allows us to investigate whether the observed group fairness patterns persist across different recommendation scenarios.

3.2 Data Preprocessing

Data preprocessing consisted of two main stages: data cleaning and transformation, followed by sampling. During data cleaning, users and items without interactions were removed. Users with missing or incomplete metadata, along with their interactions, were also excluded because these attributes are required for the user grouping procedures described in Section 3.5.

Dataset-specific preprocessing was then applied. For LastFM, artists without an MBID or artist name were removed, and duplicate user-artist interactions were aggregated by summing their play counts. Since LastFM interactions are represented as play counts, we transformed them to a 1-to-5 scale. First, each play count c was transformed using $\log(1 + c)$. The resulting values were then rescaled according to

$$r = 1 + 4 \frac{\log(1 + c) - \min}{\max - \min},$$

where min and max were computed exclusively from the global training partition. The same values were subsequently used to transform the validation and test interactions, and the resulting ratings were clipped to the $[1, 5]$ interval. This procedure prevents information from the held-out partitions from influencing the transformation.

For Yelp, we retained users whose majority of evaluated businesses were restaurants, resulting in approximately 77% of users. When multiple ratings from the same user for the same business were available, only the most recent rating was retained.

After preprocessing, we sampled each dataset using a strategy inspired by the experimental setup adopted in previous work [16].

¹<https://business.yelp.com/data/resources/open-dataset/>

Table 1: Dataset dimensions before and after the sampling procedure.

Dataset	Stage	Users	Items	Density
LastFM	Original	359,347	294,015	0.016620%
	Sampled	10,000	10,000	0.410337%
Yelp	Original	1,987,897	150,346	0.002339%
	Sampled	10,000	10,000	0.157285%

We first selected the 10,000 items with the highest number of interactions. From the users interacting with these items, we then randomly sampled 10,000 users with at least six interactions. User clustering, described in 3.5, was performed after preprocessing and sampling. This minimum was required to ensure that each user could contribute interactions to all partitions used during the cross-validation procedure described in Section 3.4. Table 1 summarizes the statistics of the original and sampled datasets.

3.3 Recommendation Models

We evaluated two neural recommendation models with distinct architectures: NeuMF and MultiVAE. These models were selected to investigate group fairness under different neural approaches to collaborative filtering.

- **NeuMF** (*Neural Collaborative Filtering*): This model combines Generalized Matrix Factorization (GMF) and a Multi-Layer Perceptron (MLP) to model user-item interactions. While the GMF component captures linear interaction patterns through latent representations, the MLP component learns non-linear relationships between users and items. Their representations are combined to produce the final recommendation score [9].
- **MultiVAE** (*Variational Autoencoders for Collaborative Filtering*): This model is a variational autoencoder designed for collaborative filtering. It encodes a user's interaction vector into a latent representation and reconstructs the distribution over items from this representation. The model uses a multinomial likelihood objective to learn user preferences from interaction data [13].

The comparison between NeuMF and MultiVAE allows us to evaluate whether group fairness patterns remain consistent across two neural recommendation architectures that learn user preferences through different mechanisms.

3.4 Hyperparameter Optimization

Hyperparameter optimization was performed using 5-fold cross-validation within the global training partition. First, for each user, 80% of the interactions were assigned to the global training set, while the remaining 20% were held out as the test set and were not used during hyperparameter selection.

The interactions in the global training set were then distributed across five folds. At each cross-validation iteration, four folds were used for training and the remaining fold for validation. Therefore, each hyperparameter configuration was trained and evaluated five times, with each fold serving as the validation set exactly once.

Relative to the complete dataset, each iteration used approximately 64% of the interactions for training, 16% for validation, and kept the same 20% as the held-out test set.

Each hyperparameter configuration was evaluated using NDCG@10, and the configuration with the highest mean NDCG@10 across the five validation folds was selected for each recommendation model. Hyperparameter selection was therefore based exclusively on recommendation performance, independently of the group fairness measurements analyzed afterward.

For NeuMF, the evaluated hyperparameters were: learning rate $\in \{0.0005, 0.001\}$, MLP hidden layers $\in \{[64, 32, 16], [128, 64]\}$, MF embedding size $\in \{32, 64\}$, and MLP embedding size $\in \{32, 64\}$. For MultiVAE, we evaluated learning rate $\in \{0.0005, 0.001\}$, dropout probability $\in \{0.3, 0.5\}$, and latent dimension $\in \{128, 200\}$.

After selecting the best hyperparameter configuration, the corresponding model was retrained using the entire global training set and evaluated on the held-out test set.

3.5 User Grouping

Users were analyzed under two grouping approaches: predefined groups based on observable user attributes and latent groups identified through unsupervised clustering. For LastFM, the observable attributes were age, gender, location, and activity. For Yelp, we considered number of friends, number of fans, tenure, and activity. Latent groups were identified using K-Means and Agglomerative Clustering in both datasets.

The predefined groups were organized as follows:

- Activity: users whose number of interactions was above the 95th percentile were classified as active, while the remaining users were classified as inactive.
- Age: users were divided into seven groups: under 18, 18–24, 25–34, 35–44, 45–49, 50–55, and over 55.
- Gender: users were divided into male and female groups.
- Location: users were divided into three geographic groups: North America, comprising users from the United States and Canada; Europe, comprising users from European countries; and Other Locations, comprising users from all remaining countries in the dataset, including countries in Central and South America, Asia, Oceania, and Africa.
- Friends: users were divided into four groups according to their number of friends: no friends, 1–10, 11–100, and 101 or more. These intervals were manually defined after inspecting the empirical distribution of friend counts, with the objective of separating users with no social connections, low and intermediate connectivity, and highly connected users.
- Fans: users were divided into four groups according to their number of fans: 0–1, 2–7, 8–13, and 14 or more. The intervals were manually defined through visual inspection of the empirical cumulative distribution. Users with 0–1 fans represent the lower region of the distribution, the two intermediate intervals represent its central portion, and users with 14 or more fans represent its upper tail.
- Tenure: users were divided into four groups according to their account tenure: less than 1 year, 1 to less than 3 years, 3 to less than 5 years, and 5 years or more.

Table 2: User grouping configurations and group composition, using the LastFM dataset.

Grouping	Groups	Quantity
Gender	Male	7392
	Female	2608
Age	Under 18	804
	18 to 24	5234
	25 to 34	2908
	35 to 44	711
	45 to 49	120
	50 to 55	86
	Over 55	137
Location	North America	2022
	Europe	6288
	Other Locations	1690
Activity	Active	515
	Inactive	9485
K-means	Group 1	515
	Group 2	9485
Agglomerative	Group 1	511
	Group 2	9489

Tables 2 and 3 present the number of users in each predefined group for the LastFM and Yelp datasets, respectively. These distributions provide an overview of the group composition in each dataset and help characterize the different user grouping configurations considered in the analysis.

In addition to the predefined groups, we applied K-Means and Agglomerative Clustering to identify latent user groups using the same metadata considered in the predefined group analysis. Numeric features were standardized using Z-score normalization; for Yelp, a *log1p* transformation was applied beforehand. Categorical features were encoded using one-hot encoding. The method employed the Euclidean distance (*metric='euclidean'*) for both algorithms. K-Means was configured with Lloyd's algorithm (*algorithm='lloyd'*), while Agglomerative Clustering used Ward's linkage criterion (*linkage='ward'*). For each method, the optimal number of clusters was selected independently based on the highest Silhouette Coefficient.

3.6 Group Fairness Evaluation

Group fairness was evaluated following the methodology adopted by [4], based on the fairness formulation proposed by [16]. After hyperparameter selection, each recommendation model was trained on the complete global training set and predictions were generated for the held-out test set.

Let X denote the matrix of observed test interactions and $\hat{X}$ the corresponding predictions generated by a recommendation model. For a user group g , let Ω_g denote the set of observed test interactions associated with users belonging to that group. The prediction loss of group g is defined as its mean squared error:

Table 3: User grouping configurations and group composition, using the Yelp dataset.

Grouping	Groups	Quantity
Activity	Active	3045
	Inactive	6955
Friends	No friends	1648
	1 to 10	2230
	11 to 100	3218
	101+	2904
Fans	0 to 1	5481
	2 to 7	2550
	8 to 13	643
	14+	1326
Tenure	Under 1 year	34
	1 to 3 years	241
	3 to 5 years	750
	5+ years	8975
K-means	Group 1	7020
	Group 2	2980
Agglomerative	Group 1	6955
	Group 2	3045

$$L_g = \frac{\|P_{\Omega_g}(\hat{X} - X)\|_2^2}{|\Omega_g|},$$

where P_{Ω_g} restricts the prediction error to the interactions associated with group g . Thus, L_g represents the average prediction error experienced by users belonging to that group.

Given G groups, group unfairness R_{grp} is computed from the pairwise differences between their losses:

$$R_{grp} = \frac{1}{G^2} \sum_{k=1}^G \sum_{l>k}^G (L_k - L_l)^2.$$

Lower values of R_{grp} indicate more similar prediction losses across groups, while larger values indicate greater disparities in recommendation error. An R_{grp} equal to zero indicates that all groups have identical prediction losses. In this work, the metric is computed independently for each user grouping described in Section 3.5.

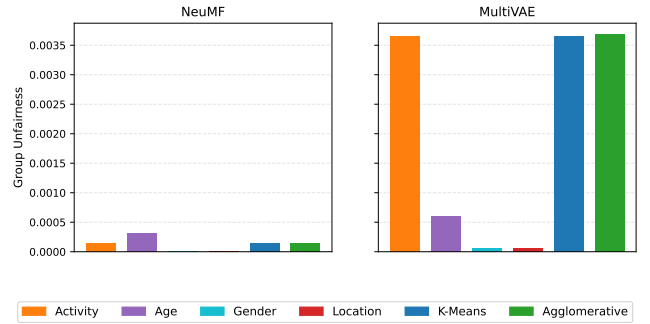
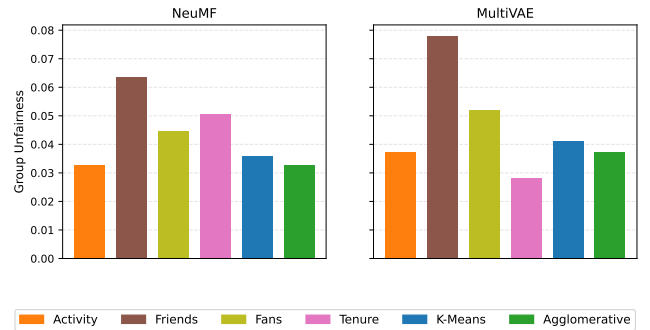
In addition to group unfairness, overall recommendation performance was measured using NDCG@10 on the held-out test set. NDCG@10 evaluates the quality of the top-ranked recommendations, whereas R_{grp} captures how unevenly prediction errors are distributed across user groups.

4 Results and Discussion

This section analyzes group fairness across the NeuMF and MultiVAE recommendation models under different user grouping configurations. The experiments were conducted on LastFM and Yelp, which represent distinct recommendation domains and forms of user feedback. Figures 1 and 2 present the group unfairness (R_{grp})

obtained for each grouping configuration, while Figures 3 and 4 show the corresponding group losses (L_g). Figures 5 and 6 summarize the distribution of unfairness values across the evaluated groupings. Tables 4 and 5 complement these results with the R_{grp} values and the NDCG@10 obtained by each recommendation model. The following analysis addresses RQ1 by examining disparities across predefined user groups and RQ2 by comparing these results with the clustering-based groupings.

The LastFM results in Figure 1 show distinct fairness patterns between NeuMF and MultiVAE. For NeuMF, Age produced the highest group unfairness ($R_{grp} = 0.00031714$), while the remaining configurations resulted in considerably lower disparities. For MultiVAE, the largest values were obtained for Activity, K-Means, and Agglomerative Clustering, with R_{grp} values around 0.0037. In particular, Activity and K-Means resulted in exactly the same R_{grp} , while Agglomerative produced a very similar value.

**Figure 1: Analysis of group unfairness (R_{grp}) by recommendation model and user grouping using the LastFM dataset.****Figure 2: Analysis of group unfairness (R_{grp}) by recommendation model and user grouping using the Yelp dataset.**

These similarities are consistent with the group compositions presented in Table 2. Activity separates 515 active users from 9,485 inactive users, while K-Means produces groups containing exactly 515 and 9,485 users. Agglomerative Clustering generates a similarly imbalanced partition, with 511 and 9,489 users. Together with the similar group losses shown in Figure 1, these results indicate that, in LastFM, the clustering-based partitions closely resemble the

structure observed for user activity rather than exposing a clearly different source of disparity.

A different pattern is observed for Yelp. As shown in Figure 2, Friends produced the highest group unfairness for both NeuMF and MultiVAE, reaching 0.06349989 and 0.07795636, respectively. Fans also resulted in relatively high disparities, whereas the clustering-based configurations did not exceed the unfairness observed for these predefined groups. MultiVAE presented higher R_{grp} than NeuMF for all grouping configurations except Tenure, indicating that the fairness differences between recommendation models also depend on the user characteristic used to define the groups.

The Yelp clustering results again show an important relationship with Activity. Agglomerative Clustering divides the users into groups of 6,955 and 3,045 users, which exactly matches the sizes of the inactive and active groups, respectively. Moreover, both configurations produce the same R_{grp} for each recommendation model. K-Means also generates a similar partition, although with slightly different group sizes. Therefore, as in LastFM, the clustering-based partitions closely resemble the Activity grouping at the aggregate level. However, identical group sizes and R_{grp} values alone do not establish that the partitions contain exactly the same users.

This result contrasts with the behavior reported in the previous study [4], where Agglomerative Clustering frequently exposed larger disparities than predefined user groups. Under the experimental conditions evaluated in this work, clustering did not consistently reveal groups with higher unfairness. Instead, observable characteristics such as Age in LastFM and Friends in Yelp were sometimes more informative for identifying disparities. These findings suggest that the relevance of latent groups is dependent on the dataset, the available user attributes, and the recommendation model, rather than being a systematic property of the clustering method.

Figures 3 and 4 provide further insight into the disparities summarized by R_{grp} . In LastFM, active users experience higher prediction losses than inactive users for both recommendation models. Age also reveals substantial differences, particularly for users under 18 and, for MultiVAE, users over 55. In contrast, Gender and Location present relatively similar losses across their respective groups, which is consistent with their low R_{grp} values.

For Yelp, Figure 4 shows a clearer relationship between user characteristics and prediction error. Users with no friends exhibit the highest loss within the Friends configuration, while the loss progressively decreases for users with larger social networks. Similar differences appear for Fans, where users with fewer fans receive higher prediction errors. Activity follows the same general pattern, with inactive users experiencing higher losses than active users. These results indicate that behavioral and social characteristics can be associated with substantial differences in recommendation quality, even when they are not the attributes recovered most strongly by the clustering procedures. These results further highlight activity as a relevant grouping criterion, consistent with previous evidence that recommendation quality and fairness can differ across users with different activity levels [15].

Tables 4 and 5 also show that recommendation performance and group fairness do not necessarily vary in the same direction. In Yelp, MultiVAE achieves a higher NDCG@10 than NeuMF, but also produces higher group unfairness in five of the six evaluated grouping configurations. Conversely, in LastFM, NeuMF achieves the higher

Table 4: Comparison of group unfairness (R_{grp}) and NDCG@10 across recommendation models and user grouping criteria—using the LastFM dataset.

Model	Grouping	R_{grp}	NDCG@10
NeuMF	Activity	0.00014539	0.0084611
	Age	0.00031714	
	Gender	0.00000001	
	Location	0.00001310	
	K-Means	0.00014539	
	Agglomerative	0.00014794	
MultiVAE	Activity	0.00365356	0.0011623
	Age	0.00060994	
	Gender	0.00005595	
	Location	0.00005495	
	K-Means	0.00365356	
	Agglomerative	0.00368889	

Table 5: Comparison of group unfairness (R_{grp}) and NDCG@10 across recommendation models and user grouping criteria—using the Yelp dataset.

Model	Grouping	R_{grp}	NDCG@10
NeuMF	Activity	0.03275497	0.0006963
	Friends	0.06349989	
	Fans	0.04468568	
	Tenure	0.05074751	
	K-Means	0.03608613	
	Agglomerative	0.03275497	
MultiVAE	Activity	0.03733713	0.0014222
	Friends	0.07795636	
	Fans	0.05214720	
	Tenure	0.02812023	
	K-Means	0.04111847	
	Agglomerative	0.03733713	

NDCG@10 and substantially lower unfairness for Activity and the clustering-based configurations. These observations reinforce the importance of evaluating recommendation quality and group fairness separately, since an improvement in ranking performance does not necessarily imply a more uniform distribution of prediction error across user groups. This distinction is consistent with broader research showing that traditional recommendation objectives alone do not capture all fairness-related considerations in recommender systems [1].

Finally, Figures 5 and 6 summarize the variability of R_{grp} across grouping configurations. In LastFM, MultiVAE exhibits substantially greater dispersion because Activity and the two clustering configurations produce unfairness values considerably higher than those obtained for Gender, Location, and Age. NeuMF remains comparatively stable, with Age appearing as an isolated higher value. In Yelp, both models show greater variation across grouping criteria,

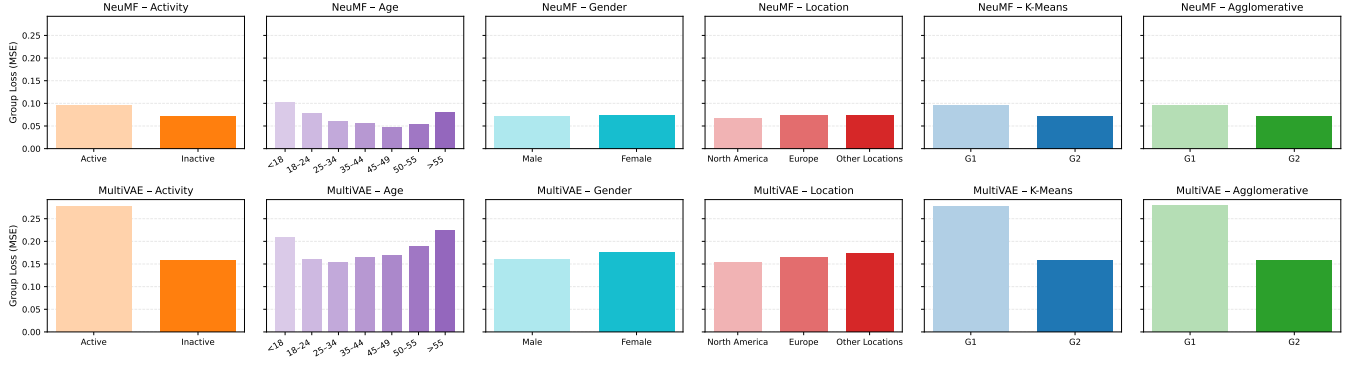


Figure 3: Group loss values (L_g) by recommendation model and user grouping using the LastFM dataset.

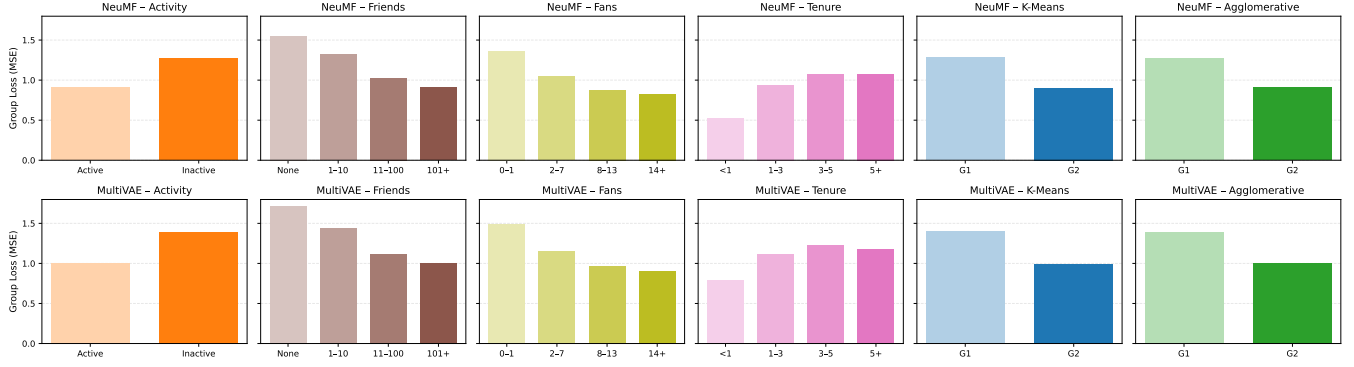


Figure 4: Group loss values (L_g) by recommendation model and user grouping using the Yelp dataset.

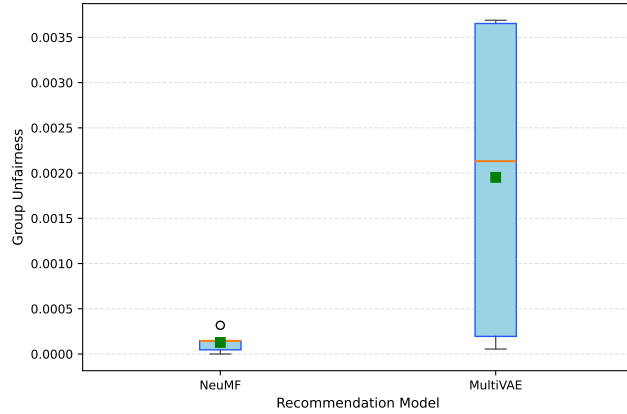


Figure 5: Distribution of group unfairness across recommendation models using the LastFM dataset.

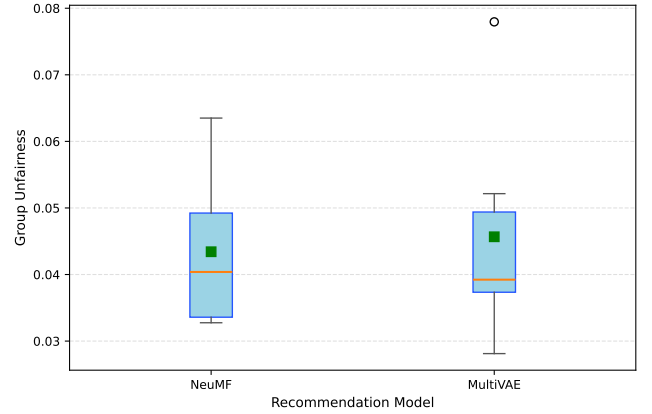


Figure 6: Distribution of group unfairness across recommendation models using the Yelp dataset.

with Friends producing the highest unfairness and Tenure producing the lowest value for MultiVAE. Overall, these distributions further demonstrate that conclusions about fairness can change substantially depending on how the user population is partitioned.

Taken together, the experiments do not indicate that latent clustering systematically reveals the most unfair user groups. Instead,

the results show that disparities may be associated with either predefined or data-driven groups depending on the dataset and recommendation model. Moreover, the strong correspondence between the clustering solutions and Activity in both datasets suggests that inspecting the composition of latent groups is essential

before interpreting them as evidence of previously hidden fairness disparities.

5 Conclusions

The analysis of the results provides further evidence that group fairness in recommender systems depends on the interaction between the recommendation model, the dataset, and the way users are grouped. Across LastFM and Yelp, NeuMF and MultiVAE exhibited different levels of group unfairness depending on the grouping criterion, showing that similar recommendation strategies may affect user groups differently.

Unlike previous results obtained with traditional collaborative filtering methods, the latent groups identified through K-Means and Agglomerative Clustering did not systematically expose the highest levels of unfairness. In both datasets, the clustering solutions were strongly associated with user activity, frequently reproducing or closely approximating the corresponding predefined groups. At the same time, observable characteristics such as age in LastFM and number of friends in Yelp revealed larger disparities in some experimental configurations.

These findings highlight the importance of examining the composition of data-driven groups before interpreting them as evidence of hidden unfairness. Clustering can provide an additional perspective for fairness analysis, but the resulting groups may reflect structures that are already captured by observable user characteristics. Therefore, fairness evaluation should consider multiple grouping criteria together with recommendation performance rather than relying on a single predefined or clustering-based partition.

As future work, additional clustering strategies and user representations could be investigated to determine whether alternative definitions of latent groups reveal disparities that are not dominated by activity-related characteristics. Extending the analysis to other recommendation models and datasets, as well as evaluating fairness mitigation strategies for the identified groups, may also contribute to a broader understanding of group unfairness in recommender systems.

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